

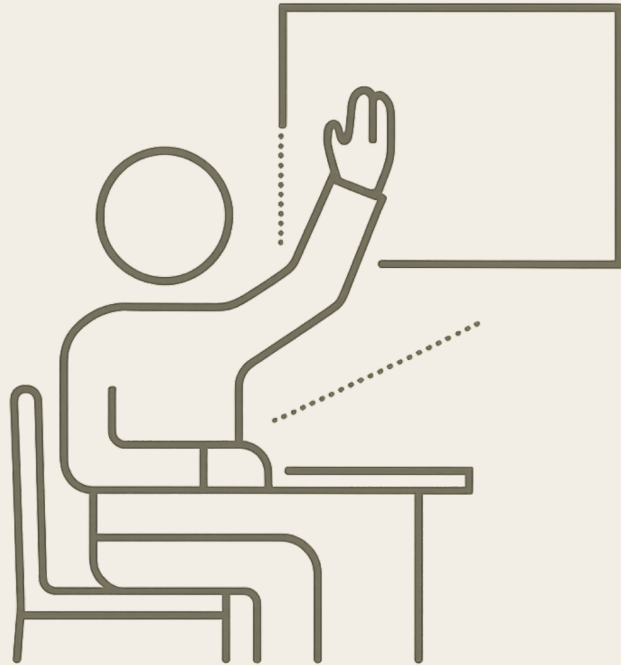


GUEST LECTURE · DAYALBAGH EDUCATIONAL INSTITUTE

Careers in the Age of Data & AI

Snehith Allamraju

Senior Director, Data Analytics · Dun & Bradstreet



So What's Left for You?

Hands up if you've wondered whether AI will take your job.

It's the question on every engineering student's mind right now — and it deserves an honest answer. Tonight, we're going to tackle it head-on, with real data, a live demo, and a career map that shows exactly where you fit in this new world.

19+ YEARS IN DATA ANALYTICS & AI

My Path in Three Turning Points

1

Turning Point 1 · Infosys

The decision that set the direction — choosing data over a more conventional engineering path when most peers were chasing other streams.

2

Turning Point 2 · Nobel Biocare / Envista

A setback that became a master class in resilience. Moving into a global healthcare analytics role forced a steep, uncomfortable learning curve — and it was worth every bit of it.

3

Turning Point 3 · RSM US to D&B

Built and led a 30+ person India data & AI team at RSM. Then: Senior Director at Dun & Bradstreet in 2026 — proof that every compounded year of learning changes what becomes possible.

Why Analytics? Why Now?

India's digital economy is scaling at an unprecedented rate. For 3rd and 4th year engineering students, this structural talent shortfall means your hiring runway is wide open — if you build the right skills today.

2117+

GCCs in India

Operating in FY26, collectively employing 2.36M professionals across functions.

4.5L+

New GCC Jobs

Projected for 2026 — the majority skewed toward analytics and data roles.

20–23%

Hyderabad's Share

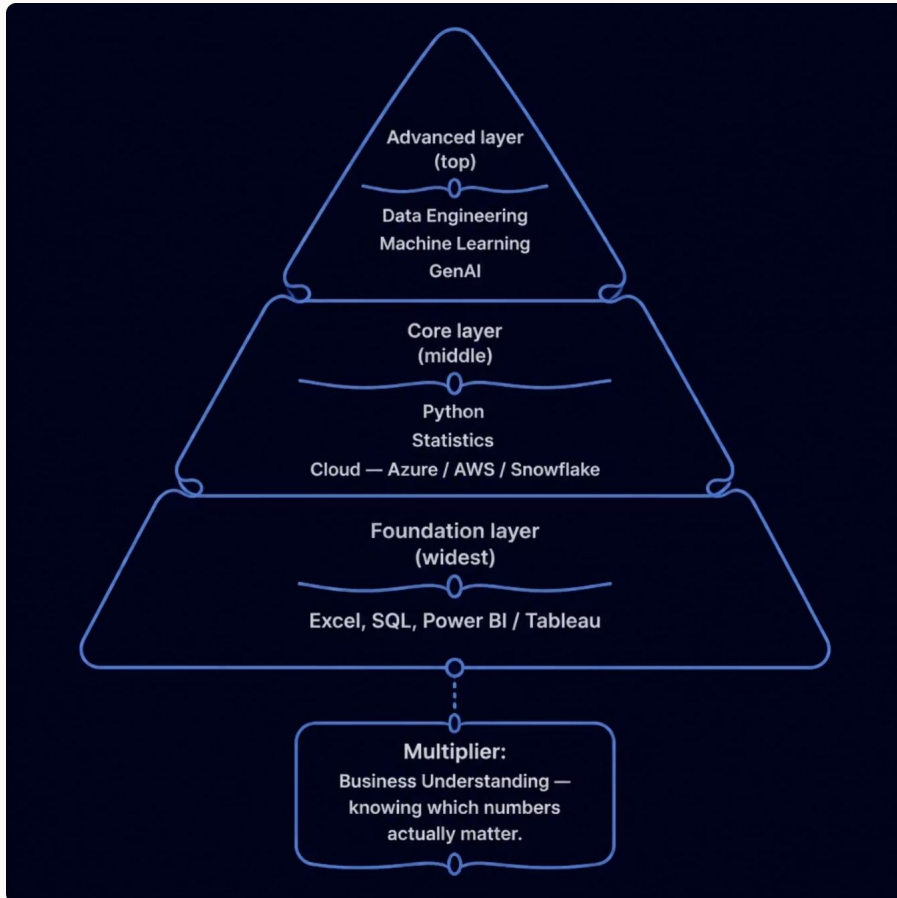
Of India's entire GCC footprint, with a strong BFSI & analytics concentration.

200K+

Talent Shortfall

Analytics practitioners India's digital economy urgently needs right now.

The Analytics Skill Pyramid



Build from the bottom up. Every advanced skill rests on a strong foundation. Don't skip ahead — the pyramid collapses without it.

Foundation Layer

Excel · SQL · Power BI / Tableau — the non-negotiables every employer checks first.

Core Layer

Python · Statistics · Cloud (Azure / AWS / Snowflake) — tools that give you analytical horsepower.

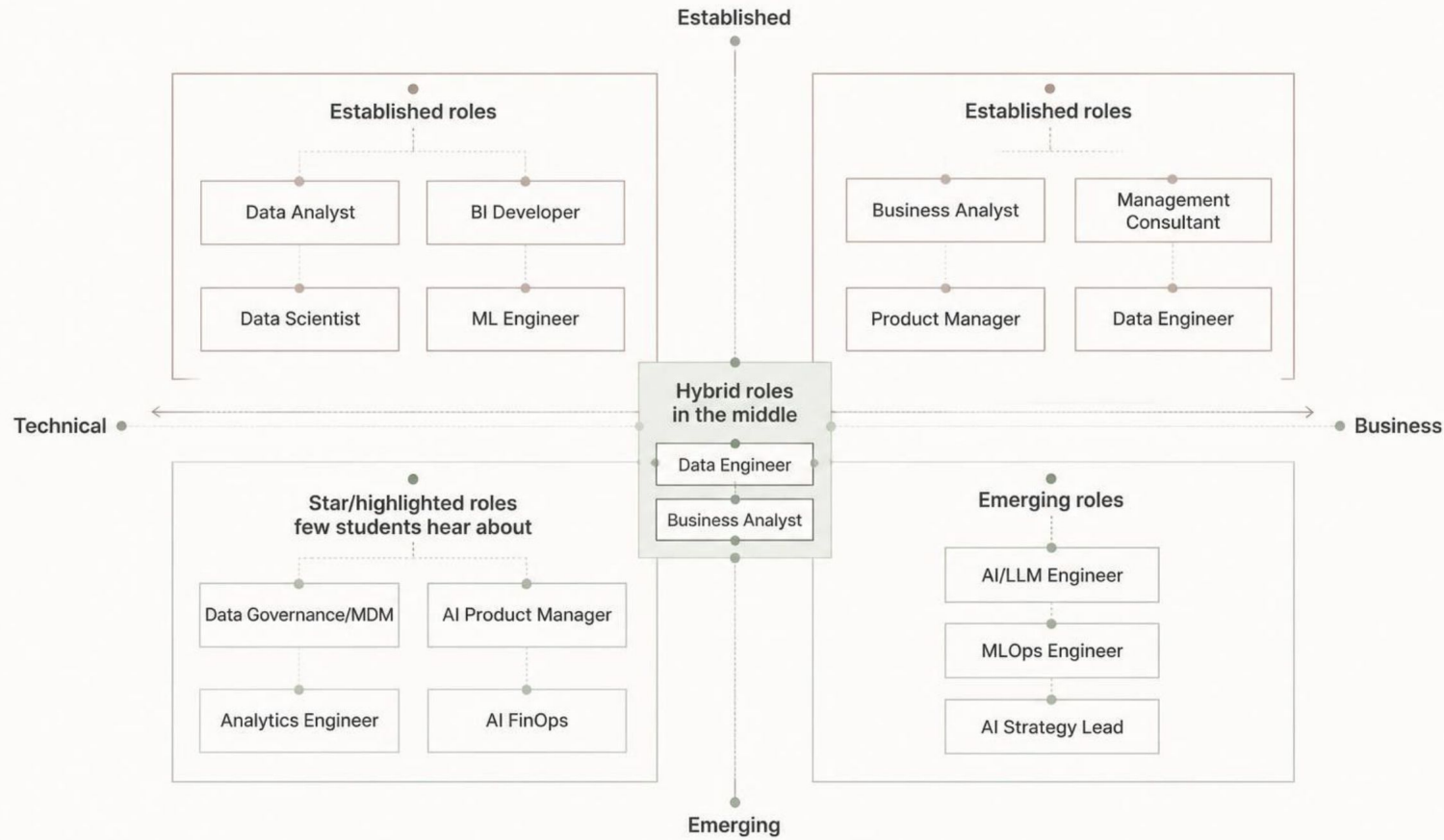
Advanced Layer

Data Engineering · Machine Learning · GenAI — the top of the market.

❏ **The multiplier at every level:** Business Understanding — knowing which numbers actually matter to your company.

Where Will You Fit? The Data & AI Career Map

The analytics landscape has expanded far beyond "Data Analyst." Roles that barely existed three years ago are now among the highest-paying in GCCs. Here's the full map — technical paths, business paths, and the emerging roles that most students never hear about.



★ **Roles few students hear about** — Data Governance / MDM, AI Product Manager, Analytics Engineer, AI FinOps — carry premium pay and very low competition from fresh graduates right now.

"Will AI Replace Data Analysts?"

No.

AI replaces **repetitive work**. It amplifies **capable professionals**.

What AI genuinely cannot replace:

- **Business context** — knowing which numbers actually matter to this company, this quarter
- **Stakeholder trust** — getting a sceptical VP to act on your analysis
- **Framing ambiguity** — turning "look into this" into a solvable question
- **Accountability & data ethics** — owning outcomes when the model is wrong

❏ Future employers want **AI-enabled analysts** who use Copilot, ChatGPT, build agents, and automate reporting — not analysts who fear AI.

What AI Takes Over — and What Compounds

Losing Value Fast

- Writing routine SQL and reports by hand
- Cleaning data one tedious step at a time
- Memorising tool syntax rather than understanding concepts

Compounds Over a Career

- Framing the right question before touching the data
- Knowing how a business really works, not just how the database works
- Judging whether the data can actually be trusted for this decision

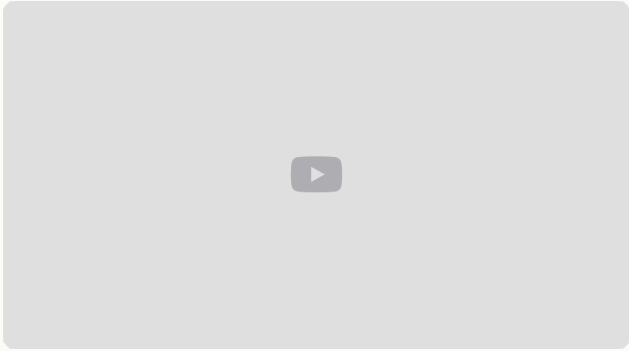
The professionals who thrive are not those who resist AI — they are those who learn to direct it, question its outputs, and translate its answers into business decisions that matter.

OPENING CONTEXT

Every Order Leaves a Trace.

A food-delivery business in Agra. 3,400 orders. One messy file. Eight roles. One burning question. What follows is a live walkthrough of how every data and AI role on the career map touches a single, real-world business problem — together, in a relay.







LIVE DEMO · ONE QUESTION, EIGHT ROLES

Why Are Repeat Orders Dropping?

A food-delivery business in Agra. 3,400 orders. One messy, real-world file — exactly the kind you will encounter on Day 1 of any analytics job. Over the next few slides, eight different roles each add one layer of value to answering this single business question. Watch how the answer changes with each lens.

Eight Roles, One Relay

01

Business Analyst

Frames the question. Defines what "dropping" actually means and which metric to move.

03

Data Governance / MDM

Turns cleanup into tested, trusted models. One real restaurant, one canonical record.

05

Analytics Engineer

Turns the analyst's logic into reusable, version-controlled dbt models for the whole team.

07

AI FinOps

Tracks what every AI query costs. Keeps cloud spend aligned with business value.

02

Data Engineer

Makes messy raw data usable. Ingests, normalises, and pipelines the 3,400-order file.

04

Data Analyst

Finds the insight worth acting on — late deliveries, reorder rates, area-level patterns.

06

AI / LLM Engineer

Lets anyone ask questions in plain English — AI writes the SQL and returns the answer.

08

AI Product Manager

Decides what to build next — which AI feature actually solves the business problem.

Day One: The Raw File

Real data never arrives clean. Before any analysis, modelling, or AI can happen, someone must tame the chaos. That someone is the Data Engineer — and this is what they find on Day 1.

3,400

Orders

Total records in the raw delivery file — each one a potential insight waiting to be unlocked.

3

Date Formats

Three different date formats used across the same column. All must be standardised before any time-series analysis.

24

Spellings

Twenty-four unique spellings for just 8 delivery areas. Duplicates that silently distort every downstream report.

"27 min"

Text, Not Number

Delivery time stored as text with ₹ symbols embedded in amount fields — cannot be summed or averaged as-is.

Which Restaurant Earns the Most?

Master Data Management is the unglamorous work that makes every other analysis trustworthy. Without it, you are drawing conclusions from a broken mirror.

Raw Data Says

#1 Agra Tandoor House

~₹1.8 lakh in revenue — because only one spelling is counted.

After Merging Duplicates

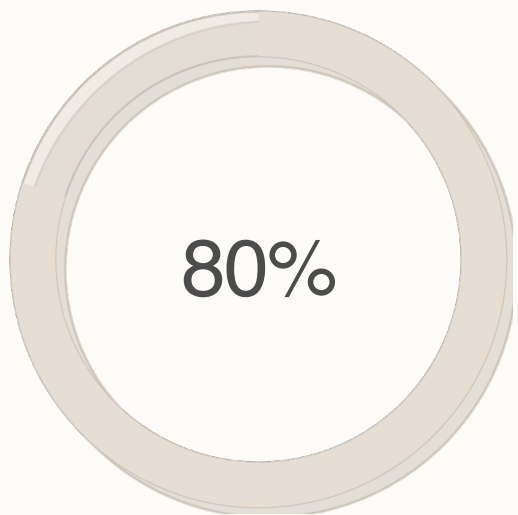
#1 Mama's Kitchen

~₹2.8 lakh in revenue — **four spellings, one restaurant**. The leaderboard changes completely once the data is trusted.

- ❏ Business decisions made on uncleaned data are not just inaccurate — they can be actively harmful. Data governance is what separates a trustworthy analytics function from a dangerous one.

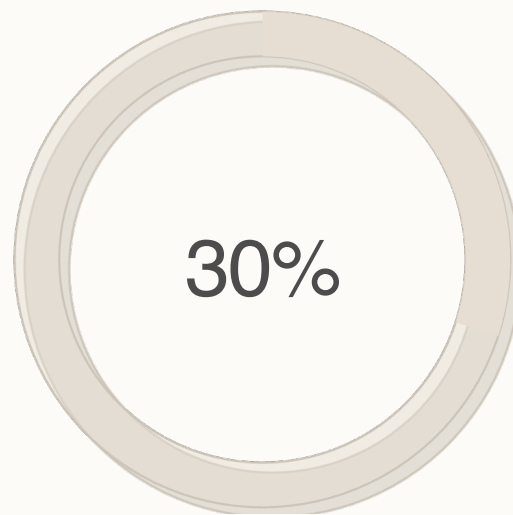
Late Deliveries Lose Customers

The analyst's job is to find the one number that changes behaviour. Here it is — and it is stark.



Reorder — On Time

8 in 10 customers reorder after a delivery arrives on time. Reliability is the product.



Reorder — Late

Only 3 in 10 return after a late delivery. A single bad experience cuts repeat revenue by more than half.

 **Sikandra and Shahganj:** nearly 1 in 2 deliveries arrive late. Everywhere else in the dataset: about 2% late. The fix is geographic, not product-level.

Every AI Answer Has a Bill



The AI / LLM Engineer makes data accessible to non-technical stakeholders — anyone can type a question in plain English and get an instant SQL-backed answer. The AI FinOps role then tracks the cost of every query: tokens consumed multiplied by the cloud model's price per million tokens. Intelligent organisations optimise for both insight quality and infrastructure spend.

What Recruiters Actually Look For

GCC hiring managers are not checking your CGPA first. They are looking for evidence that you can actually do the work. Here are the four signals that open doors.



SQL Fluency

Can you pull and shape data unassisted — not just recite syntax from memory? Expect a live SQL test in most data interviews.



Communication

Can you explain an insight to a non-technical executive in under 2 minutes? This is the skill that separates the shortlisted from the hired.



Proof of Work

A working project beats a stack of certificates every single time. Show what you can actually build and publish it where recruiters can open it.



Curiosity & Problem-Solving

Resume + Projects + GitHub + LinkedIn + attitude. The candidate who asks "why?" and then goes and finds out wins. Not CGPA alone.

Biggest Mistakes Students Make

✗ Breadth Over Depth

Learning 20 tools instead of mastering the fundamentals. SQL and Python done well beats a résumé listing 15 half-learned tools. Go deep first.

✗ Certificates Without Projects

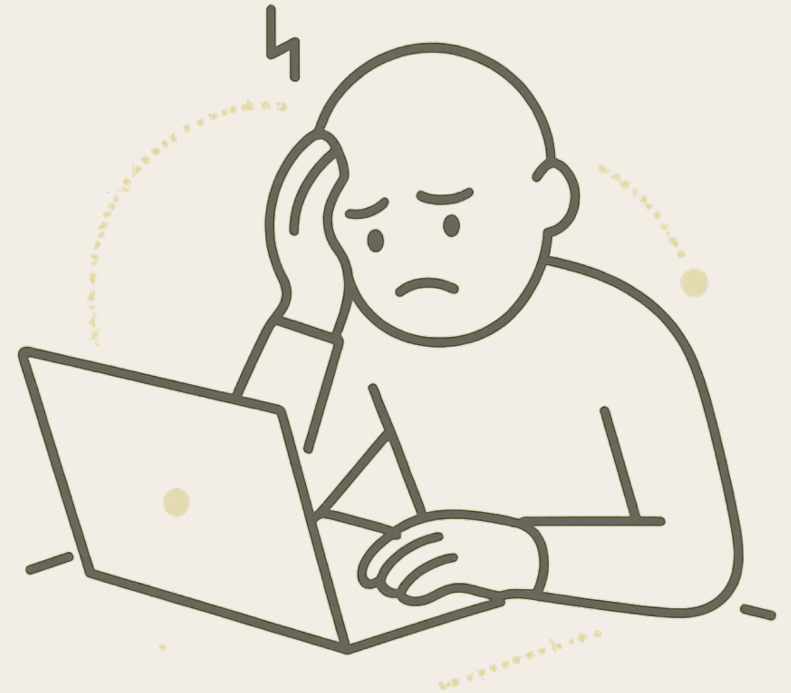
Collecting badges with zero finished, end-to-end projects that recruiters can actually open, run, and evaluate. Badges signal intent; projects prove capability.

✗ Zero Visibility

No LinkedIn, no GitHub, no public work. If a recruiter cannot find you online, you functionally do not exist in their hiring pipeline.

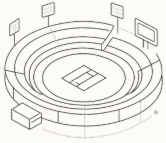
✗ Starting Too Late

Waiting until final year to begin. The best placements go to students who started building 12–18 months before placement season opens.



Projects That Impress Recruiters

The best projects share one thing: they go from raw data to a clear "so what?" — not just a model score. Publish it. A repo someone can open and a dashboard you can walk through live in 5 minutes beats any badge.



IPL Analytics

Batting form, win predictors, team performance over seasons. Familiar data, high engagement from interviewers who are fans.



Netflix Dashboard

Content trends, genre popularity, ratings analysis. Demonstrates end-to-end BI skills with a globally recognised dataset.



Uber / Ride Data

Surge pricing patterns, demand hotspots, driver efficiency. Real-world operational analytics that mirrors GCC project work.



Fraud Detection

Imbalanced classification, precision-recall trade-offs. Signals ML maturity and business awareness simultaneously.



Spotify Analysis

Audio features, playlist curation, listening patterns. Creative domain with rich public APIs — shows initiative.

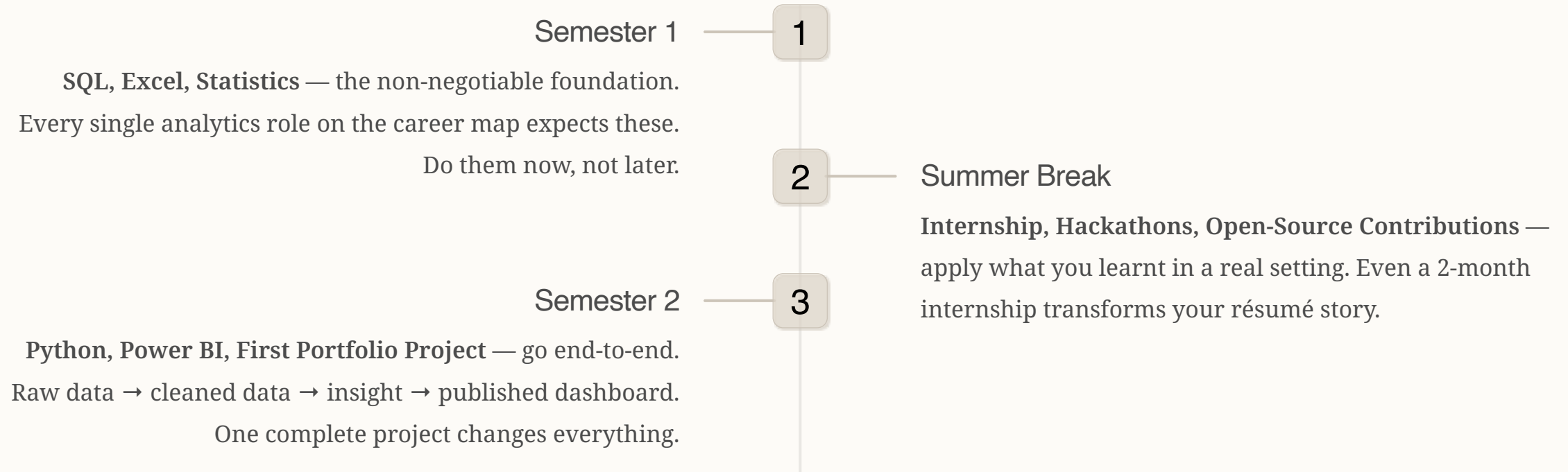


GenAI Chatbot / RAG

A retrieval-augmented generation system on any document set. Immediate signal that you are AI-enabled, not AI-afraid.

Roadmap: 3rd Year

The 3rd year is your most important preparation window. Build the foundation now — SQL, Python, and one solid project — so that your 4th year is spent polishing and placing, not starting from scratch.



Roadmap: 4th Year

1 Polish Your Presence

Resume · LinkedIn · GitHub — make every piece of your work visible to recruiters. A polished profile is the lowest-effort, highest-leverage move you can make.

3 Build AI Skills

Copilot, ChatGPT prompting, automation pipelines — be the AI-enabled candidate who uses these tools to do better work faster, not the one who is scared of them.

2 Practice Out Loud

Mock interviews · SQL case questions spoken aloud, not just typed on paper · Storytelling with data. Communication under pressure is a learnable skill — practise it deliberately.

4 Network Intentionally

Meetups · Hackathons · Community events — show up, volunteer, speak. Titles matter less than learning. Every role gives the next one credibility. Stay curious and keep building even after placement.

Career Growth & Salary Reality

Analytics careers in GCCs and product companies offer one of the fastest salary progression tracks in Indian tech. Every role you take compounds — the learning in year 2 makes you 3× more valuable in year 5.

Stage	Experience	CTC Range	Scope
Associate Analyst	0–1 yr	₹4–7.5 LPA	Data prep, basic SQL dashboards, ad-hoc reporting
Senior Analyst	2–5 yr	₹8–14 LPA	Automation scripts, model ownership, cross-functional stakeholders
Lead / Manager	5–8 yr	₹12–20 LPA	Strategic partnering, guiding junior analysts, programme ownership
Director / CDO	10+ yr	₹25 LPA+	Enterprise data architecture, organisation-wide data strategy

❏ These are market ranges, not ceilings. Skills, domain expertise, and the ability to communicate with leadership are the levers that push you to the top of each band.

My Favourite Resources

Most of the best learning in analytics is free or very low cost. Here is the stack I would use if I were starting today — no expensive bootcamps required.



SQL Practice

LeetCode SQL · Mode Analytics SQL — solve problems daily, not just in the week before an interview. Consistency beats cramming every time.



Projects & Real Data

Kaggle · GitHub — real datasets, real projects, real public visibility. Kaggle competitions also build a track record recruiters can see.



Learning Platforms

Microsoft Learn · Snowflake University · Coursera · YouTube — structured paths and free certifications. Most of the best content costs nothing.



Community

Show up to meetups, volunteer at hackathons, speak when you get the chance. Relationships accelerate careers faster than any certificate.

Lessons I Wish Someone Had Told Me

Learn SQL Early. Use It Forever.

I have used SQL every single year of my 19-year career. Not occasionally — every year, in every role, at every level. Start now. It is the one skill that never becomes irrelevant.

Chase Learning, Not Salary.

The money follows when you compound your skills relentlessly, year after year. Every role that stretches you is an investment. Every comfortable role that doesn't is a cost.

Communication Separates Good from Great.

Technical skill gets you into the room. Communication determines what happens once you are in it. Learn to tell the story behind the number — that is where careers accelerate.

Technology Changes. Curiosity Doesn't.

The tools you learn today will be partly obsolete in 3–5 years. Your relationships, your mindset, and your ability to learn fast will not. Invest in those above everything else.

Your Degree Gets You the Interview. Your Skills Build the Career.

"Your degree gets you an interview. Your skills build your career. Your attitude determines how far you go."

Tonight's Challenge — Start Right Now:



Update LinkedIn

Add your projects, skills, and a professional photo tonight. Recruiters are searching right now.



Install VS Code

Set up your analytics development environment. This is your cockpit for every project ahead.



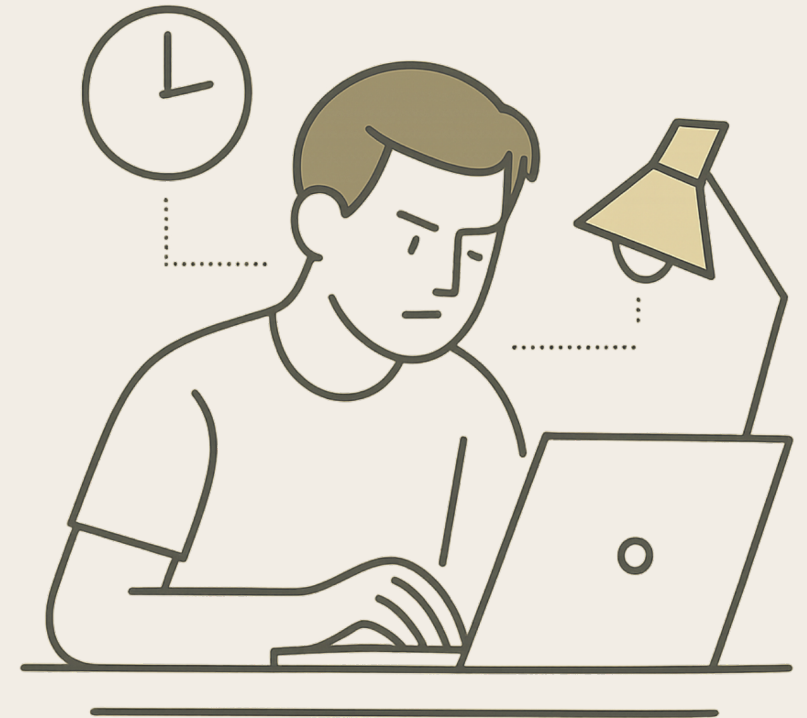
Create a GitHub Account

Your public portfolio starts with a single repository. The first commit is the hardest — make it tonight.



Solve One SQL Problem

LeetCode or Mode Analytics — pick one problem and solve it before you sleep tonight. One a day changes everything.





Snehith Allamraju

Senior Director, Data Analytics
Operations at Dun & Bradstreet



"The data world doesn't need more generalists. It needs specialists who know how to ask the right question."

Ask Better Questions. Build Faster Careers.

The single biggest competitive advantage you can build in the next 12 months is not another tool or another certificate. It is the discipline to go deep on one thing, finish one project, and ship it publicly — so that the next recruiter who searches your name finds evidence, not just a name.

Snehith Allamraju · Senior Director, Data Analytics · Dun & Bradstreet

Let's connect on LinkedIn.

Your Next 90 Days

Careers are built in 90-day sprints, not in one dramatic decision. Here is your concrete, no-excuses action plan starting the moment you leave this room tonight.

1

Days 1–30: Learn SQL

Solve 30 problems on LeetCode SQL or Mode Analytics. One per day. Every role on the career map — technical, hybrid, or business — uses SQL. This is the foundation nothing else works without.

2

Days 31–60: Go Deep on One Tool

Pick the one tool that best fits your path — Power BI, Python, dbt, or a cloud platform. Go deep, not wide. Earn one free certification to mark the milestone.

3

Days 61–90: Build in Public

One complete project published on GitHub. One LinkedIn post per month documenting what you learnt. The public track record you build now is the portfolio that gets you shortlisted.

Q&A — Your Questions, My Honest Answers

10–15 minutes open floor. Ask anything — career paths, placements, tools, salary expectations, GCCs, AI, the works. No question is too basic or too ambitious.

Guess Which Skill I've Used Every Single Year of My Career?

SQL and communication. Every. Single. Year. Across Infosys, Nobel Biocare, RSM US, and Dun & Bradstreet. If you only take two things from tonight, let it be those two.

If You Had 6 Months Before Placements, What Would You Learn First?

What would you answer? Think about it honestly for 30 seconds — then let's hear it from the room. After that, I'll share mine.

Thank you for your time and your attention tonight. Let's connect on LinkedIn and keep the conversation going.